1. The branch of mechanics that deals with the geometry of motion without considering the forces causing the motion is called:

(A) Kinematics
(B) Kinetics
(C) Statics
(D) Dynamics

2. The branch of mechanics that relates the forces acting on a body to the resulting mass and acceleration is termed:

(A) Kinetics
(B) Kinematics
(C) Statics
(D) Thermodynamics

3. In rectilinear motion, the instantaneous velocity v of a particle is mathematically defined as:

(A) v = ds / dt
(B) v = d^2s / dt^2
(C) v = Integral(s dt)
(D) v = s * t

4. The instantaneous acceleration a of a particle in rectilinear motion can be expressed in differential form as:

(A) a = dv/dt = d^2s/dt^2 = v*(dv/ds)
(B) a = s*(ds/dt)
(C) a = Integral(v dt)
(D) a = v / s

5. For a particle moving in a straight line with constant acceleration 'a', the velocity-time relationship is:

(A) v = u + a*t
(B) v = u + 0.5*a*t^2
(C) v^2 = u + a*t
(D) v = u*t + a

6. The position displacement equation for motion under uniform constant acceleration is:

(A) s = u*t + 0.5*a*t^2
(B) s = u + a*t
(C) s = 0.5*(u + v)*t^2
(D) s = u*t^2 + 2*a

7. The velocity-displacement kinematic relation independent of time for constant acceleration is:

(A) v^2 = u^2 + 2*a*s
(B) v = u + 2*a*s
(C) v^2 = u^2 + a*s^2
(D) v = sqrt(u + a*s)

8. The acceleration due to gravity 'g' on the surface of Earth in standard SI units is approximately:

(A) 9.81 m/s^2
(B) 981 m/s^2
(C) 0.981 m/s^2
(D) 98.1 m/s^2

9. When a body is projected vertically upwards under gravity, its velocity at the peak (maximum height) is:

(A) Zero
(B) Maximum
(C) 9.81 m/s
(D) Infinite

10. The time taken by a vertically projected projectile to reach maximum height from the ground with launch speed u is:

(A) t = u / g
(B) t = 2*u / g
(C) t = u^2 / (2*g)
(D) t = sqrt(2*u/g)

11. The maximum vertical height H attained by a particle projected vertically upwards with initial velocity u is:

(A) H = u^2 / (2*g)
(B) H = u / (2*g)
(C) H = 2*u^2 / g
(D) H = u^2 / g

12. In curvilinear motion described in normal-tangential (n-t) coordinates, tangential acceleration a_t represents:

(A) The time rate of change of the magnitude of velocity (speed)
(B) The rate of change of direction of velocity
(C) The total centripetal acceleration
(D) The angular displacement rate

13. In path coordinates (n-t), the normal (centripetal) acceleration a_n is given by the formula:

(A) a_n = v^2 / rho
(B) a_n = dv / dt
(C) a_n = v * rho
(D) a_n = rho / v^2

14. In the normal acceleration expression a_n = v^2 / rho, the parameter 'rho' denotes:

(A) The local radius of curvature of the trajectory
(B) The density of the moving particle
(C) The angular frequency of oscillation
(D) The friction coefficient

15. The direction of the normal acceleration vector a_n is always directed:

(A) Inward toward the center of curvature
(B) Outward away from center of curvature
(C) Tangentially along the path
(D) Opposite to the velocity vector

16. The total acceleration magnitude 'a' in normal-tangential coordinates is computed as:

(A) a = sqrt(a_t^2 + a_n^2)
(B) a = a_t + a_n
(C) a = a_t - a_n
(D) a = (a_t * a_n) / 2

17. If a particle moves along a curved path with constant speed (v = constant), its tangential acceleration a_t is:

(A) Zero
(B) Infinite
(C) v^2 / rho
(D) 9.81 m/s^2

18. If a particle travels along a straight line path (rho = infinity), its normal acceleration a_n is:

(A) Zero
(B) v^2
(C) dv/dt
(D) Infinite

19. Newton's Second Law of Motion states that the resultant force acting on a particle is proportional to:

(A) Rate of change of linear momentum
(B) Total work done
(C) Potential energy gradient
(D) Kinetic energy change

20. For a particle of constant mass m, Newton's second law is algebraically stated in vector form as:

(A) Sigma F = m * a
(B) Sigma F = m / a
(C) Sigma F = 0.5 * m * a^2
(D) Sigma F = m * v

21. The SI unit of linear momentum (p = m*v) is:

(A) kg*m/s (or N*s)
(B) kg*m/s^2 (or N)
(C) Joule (N*m)
(D) Watt (J/s)

22. D'Alembert's principle transforms a dynamic kinetics problem into an equivalent static equilibrium problem by adding:

(A) Reversed effective force (Inertia force = - m*a)
(B) Gravitational force only
(C) Centrifugal torque
(D) Damping friction force

23. The fictitious inertia force introduced in D'Alembert's dynamic equilibrium equation is directed:

(A) Directly opposite to the acceleration vector
(B) Parallel to velocity vector
(C) Toward the center of curvature
(D) Perpendicular to path

24. The work done U by a constant force F acting through displacement s at an angle theta to the motion is:

(A) U = F * s * cos(theta)
(B) U = F * s * sin(theta)
(C) U = F / (s * cos(theta))
(D) U = F * s^2

25. The SI derived unit of work and mechanical energy is the:

(A) Joule (1 J = 1 N*m)
(B) Newton (N)
(C) Watt (W)
(D) Pascal (Pa)

26. The work done by a conservative gravitational force on a mass m descending through vertical distance Delta y is:

(A) U = + m * g * Delta y
(B) U = - m * g * Delta y
(C) U = 0.5 * m * g * Delta y^2
(D) Zero

27. When a mass m is lifted vertically upwards through height h, the work done by the gravitational field is:

(A) - m * g * h
(B) + m * g * h
(C) Zero
(D) m * g / h

28. The work done in compressing or elongating a linear elastic spring of stiffness k by distance x from neutral is:

(A) U = - 0.5 * k * x^2
(B) U = - k * x
(C) U = + k * x^2
(D) U = - 2 * k * x

29. The kinetic energy (KE) of a particle of mass m translating at velocity v is defined as:

(A) KE = 0.5 * m * v^2
(B) KE = m * v
(C) KE = m * v^2
(D) KE = 0.5 * m^2 * v

30. The Work-Energy Principle for a particle asserts that the total work done by all acting forces equals:

(A) The change in kinetic energy: Sigma U_1-2 = Delta KE
(B) The total momentum change
(C) The rate of change of power
(D) The constant potential energy

31. A force field in which the total work done along any closed path is identically zero is termed a:

(A) Conservative force field
(B) Non-conservative force field
(C) Dissipative field
(D) Rotational field

32. Which of the following is a classical example of a non-conservative (dissipative) force in mechanics?

(A) Dry Coulomb sliding friction
(B) Gravitational force
(C) Linear spring restoring force
(D) Electrostatic force

33. Mechanical power P is physically defined as:

(A) The time rate of doing work: P = dW / dt = F dot v
(B) The product of work and time
(C) The integral of force over displacement
(D) The change in momentum per second

34. The SI unit of power is the Watt (W), which corresponds to:

(A) 1 Joule per second (1 J/s)
(B) 1 Newton per second (1 N/s)
(C) 1 Joule-meter
(D) 1 Pascal per second

35. The linear impulse J of a resultant force F acting over time duration Delta t is defined mathematically as:

(A) J = Integral(F dt)
(B) J = Integral(F ds)
(C) J = dF / dt
(D) J = F * v

36. The Impulse-Momentum Theorem states that the resultant linear impulse applied to a particle equals:

(A) The change in linear momentum: J = Delta p = m*v2 - m*v1
(B) The change in kinetic energy
(C) The total work performed
(D) The acceleration rate

37. If the resultant external force acting on a particle system in a given direction is zero, then:

(A) Total linear momentum in that direction is conserved
(B) Kinetic energy is conserved
(C) Total displacement is zero
(D) Velocity is zero

38. In collision mechanics, the ratio of relative velocity of separation to relative velocity of approach is:

(A) Coefficient of restitution (e)
(B) Friction factor
(C) Poisson's ratio
(D) Damping ratio

39. For a perfectly elastic direct collision between two particles, the coefficient of restitution e equals:

(A) 1.0
(B) 0.0
(C) 0.5
(D) Infinity

40. For a perfectly plastic (inelastic) collision where two colliding bodies stick together after impact, e equals:

(A) 0.0
(B) 1.0
(C) 0.5
(D) -1.0

41. A civil construction elevator ascends vertically from rest with uniform acceleration a = 1.5 m/s^2 for 6 seconds. The final velocity reached is:

(A) 9.0 m/s
(B) 6.0 m/s
(C) 12.0 m/s
(D) 4.5 m/s

42. For the elevator in Q41, the vertical distance climbed during these 6 seconds of uniform acceleration is:

(A) 27.0 m
(B) 18.0 m
(C) 36.0 m
(D) 54.0 m

43. A dump truck travelling at 20 m/s is braked to a complete stop over a distance of 40 m. The uniform deceleration is:

(A) 5.0 m/s^2
(B) 4.0 m/s^2
(C) 2.5 m/s^2
(D) 10.0 m/s^2

44. The time required for the truck in Q43 to come to rest during this 40 m stopping maneuver is:

(A) 4.0 s
(B) 2.0 s
(C) 5.0 s
(D) 8.0 s

45. A ball is thrown vertically upwards with launch speed u = 29.43 m/s. The time taken to reach the apex is (take g = 9.81 m/s^2):

(A) 3.0 s
(B) 2.0 s
(C) 4.0 s
(D) 6.0 s

46. For the ball in Q45, the maximum height attained above the release point is:

(A) 44.15 m
(B) 29.43 m
(C) 88.30 m
(D) 58.86 m

47. A stone dropped into a deep foundation shaft reaches the bottom in t = 4.0 seconds. Neglecting air drag, the shaft depth is:

(A) 78.48 m
(B) 39.24 m
(C) 156.96 m
(D) 19.62 m

48. The impact velocity of the stone in Q47 upon striking the bottom of the shaft is:

(A) 39.24 m/s
(B) 19.62 m/s
(C) 78.48 m/s
(D) 9.81 m/s

49. A particle moves along a straight line such that displacement s(t) = 2*t^3 - 6*t^2 + 5 m. Its velocity when t = 2 s is:

(A) 0 m/s (particle instantaneously at rest)
(B) 12 m/s
(C) 6 m/s
(D) 24 m/s

50. For the particle in Q49, the acceleration when t = 2 s is:

(A) 12 m/s^2
(B) 0 m/s^2
(C) 6 m/s^2
(D) 24 m/s^2

51. A particle moves with variable acceleration a = 6*t m/s^2. If v(0) = 2 m/s and s(0) = 0, the velocity function v(t) is:

(A) v(t) = 3*t^2 + 2
(B) v(t) = 6*t^2 + 2
(C) v(t) = 3*t + 2
(D) v(t) = t^3 + 2

52. For the particle in Q51, the displacement function s(t) is:

(A) s(t) = t^3 + 2*t
(B) s(t) = 3*t^3 + 2*t
(C) s(t) = 1.5*t^2 + 2*t
(D) s(t) = 0.5*t^4 + 2

53. A vehicle travels on a highway circular curve of radius rho = 200 m at a constant speed of v = 20 m/s (72 km/h). Its normal acceleration is:

(A) 2.0 m/s^2
(B) 0.1 m/s^2
(C) 4.0 m/s^2
(D) 0.5 m/s^2

54. If the vehicle on the circular curve of Q53 increases speed at a rate of dv/dt = 1.5 m/s^2, the total acceleration magnitude is:

(A) 2.5 m/s^2 (sqrt(1.5^2 + 2.0^2))
(B) 3.5 m/s^2
(C) 0.5 m/s^2
(D) 1.75 m/s^2

55. A projectile launched with speed u at angle theta to the horizontal has horizontal and vertical velocity components:

(A) u_x = u*cos(theta), u_y = u*sin(theta)
(B) u_x = u*sin(theta), u_y = u*cos(theta)
(C) u_x = u*tan(theta), u_y = u
(D) u_x = u, u_y = u

56. The total time of flight T of a projectile launched over level ground with initial velocity u at angle theta is:

(A) T = (2 * u * sin(theta)) / g
(B) T = (u * sin(theta)) / g
(C) T = (u^2 * sin(2*theta)) / g
(D) T = (2 * u * cos(theta)) / g

57. The horizontal range R of a projectile over horizontal ground launched with velocity u at angle theta is:

(A) R = (u^2 * sin(2*theta)) / g
(B) R = (u * sin(2*theta)) / g
(C) R = (2 * u^2 * cos(theta)) / g
(D) R = (u^2 * sin^2(theta)) / (2*g)

58. The launch angle theta that maximizes the horizontal range of a projectile over flat terrain is:

(A) 45 degrees
(B) 30 degrees
(C) 60 degrees
(D) 90 degrees

59. A civil water jet nozzle discharges water at u = 20 m/s at an angle of 30 degrees. The maximum height achieved by the jet stream is (g=9.81):

(A) 5.10 m
(B) 10.20 m
(C) 20.39 m
(D) 2.55 m

60. For the water jet in Q59, the horizontal range achieved over flat ground is:

(A) 35.31 m
(B) 70.62 m
(C) 17.65 m
(D) 40.77 m

61. A mass m = 50 kg rests on a smooth horizontal floor. A horizontal force P = 200 N is applied. The resulting acceleration is:

(A) 4.0 m/s^2
(B) 10.0 m/s^2
(C) 0.25 m/s^2
(D) 9.81 m/s^2

62. If a constant friction force of F_f = 50 N opposes the motion of the block in Q61, the net acceleration becomes:

(A) 3.0 m/s^2
(B) 4.0 m/s^2
(C) 2.0 m/s^2
(D) 5.0 m/s^2

63. A concrete hopper of mass m = 1000 kg is hoisted vertically by a crane cable with upward acceleration a = 2.0 m/s^2. Cable tension T is (g=9.81):

(A) 11810 N (11.81 kN)
(B) 9810 N (9.81 kN)
(C) 7810 N (7.81 kN)
(D) 2000 N (2.0 kN)

64. If the crane cable lowers the 1000 kg hopper in Q63 with downward acceleration a = 2.0 m/s^2, the cable tension T is:

(A) 7810 N (7.81 kN)
(B) 11810 N (11.81 kN)
(C) 9810 N (9.81 kN)
(D) 2000 N

65. A 10 kg block slides down a 30-degree frictionless incline. The acceleration parallel to the incline surface is (g=9.81):

(A) 4.905 m/s^2 (g * sin(30))
(B) 8.50 m/s^2 (g * cos(30))
(C) 9.81 m/s^2
(D) 2.45 m/s^2

66. If the incline in Q65 has kinetic friction coefficient mu_k = 0.20, the acceleration down the plane becomes:

(A) 3.21 m/s^2 (g*(sin 30 - mu_k*cos 30))
(B) 4.91 m/s^2
(C) 1.70 m/s^2
(D) 4.25 m/s^2

67. A pile driver hammer of mass m = 500 kg drops from rest through height h = 4.0 m onto a pile. Its velocity just before impact is (g=9.81):

(A) 8.86 m/s (sqrt(2*g*h))
(B) 12.53 m/s
(C) 6.26 m/s
(D) 4.43 m/s

68. The kinetic energy of the 500 kg hammer in Q67 just prior to impact is:

(A) 19620 J (19.62 kJ)
(B) 9810 J (9.81 kJ)
(C) 39240 J (39.24 kJ)
(D) 4905 J

69. If the hammer in Q67 drives the pile Delta s = 0.05 m (5 cm) into hard soil before stopping, the average soil resistance force is:

(A) 392.4 kN
(B) 196.2 kN
(C) 784.8 kN
(D) 98.1 kN

70. A buffer spring at the end of a railway track has stiffness k = 200 kN/m. The work required to compress it by x = 0.10 m (10 cm) is:

(A) 1.0 kJ (0.5 * 200 * 0.1^2)
(B) 2.0 kJ
(C) 20.0 kJ
(D) 0.5 kJ

71. A haulage truck engine delivers a continuous power of P = 150 kW while moving at a constant speed of v = 15 m/s. The driving traction force is:

(A) 10.0 kN (P / v)
(B) 15.0 kN
(C) 2.25 kN
(D) 2250 kN

72. A bullet of mass m = 0.02 kg (20 g) travelling at 400 m/s embeds into a wooden ballistic block of mass M = 3.98 kg at rest. The post-impact velocity is:

(A) 2.0 m/s
(B) 4.0 m/s
(C) 1.0 m/s
(D) 0.5 m/s

73. For the plastic collision in Q72, the initial kinetic energy was 1600 J. The kinetic energy remaining in the system after impact is:

(A) 8.0 J
(B) 1600 J
(C) 800 J
(D) 0.0 J

74. A ball dropped from height h1 = 9.0 m rebounds from a rigid concrete slab to height h2 = 4.0 m. The coefficient of restitution e is:

(A) 0.667 (sqrt(h2/h1) = 2/3)
(B) 0.444 (h2/h1)
(C) 0.816
(D) 0.500

75. Two identical railway freight cars A and B (mass m each) collide. Car A moves at 6 m/s, Car B is stationary. If e = 0.5, velocity of B after collision is:

(A) 4.5 m/s
(B) 3.0 m/s
(C) 1.5 m/s
(D) 6.0 m/s

76. A particle moves along a plane curve described in polar coordinates. The radial acceleration component a_r is expressed as:

(A) a_r = r_ddot - r * (theta_dot)^2
(B) a_r = r_ddot + r * theta_dot
(C) a_r = 2 * r_dot * theta_dot
(D) a_r = r * theta_ddot

77. In polar coordinates (r, theta), the transverse acceleration component a_theta is given by:

(A) a_theta = r * theta_ddot + 2 * r_dot * theta_dot
(B) a_theta = r_ddot - r * (theta_dot)^2
(C) a_theta = r * theta_dot
(D) a_theta = r_dot * theta_dot

78. The term '2 * r_dot * theta_dot' arising in the transverse acceleration of rotating reference frames is known as the:

(A) Coriolis acceleration component
(B) Centripetal acceleration
(C) Tangential Euler acceleration
(D) Bernoulli acceleration

79. A robotic civil survey arm extends radially at r_dot = 0.5 m/s while rotating at constant angular rate theta_dot = 2 rad/s. When r = 1.2 m, a_r is:

(A) - 4.8 m/s^2 (0 - 1.2 * 2^2)
(B) + 4.8 m/s^2
(C) 0 m/s^2
(D) - 2.4 m/s^2

80. For the survey arm in Q79, the transverse acceleration a_theta at that same instant is:

(A) 2.0 m/s^2 (2 * 0.5 * 2.0)
(B) 1.0 m/s^2
(C) 4.0 m/s^2
(D) 0 m/s^2

81. A particle moves such that its acceleration is proportional to displacement but directed toward the origin: a = - omega^2 * s. The resulting motion is:

(A) Simple Harmonic Motion (SHM)
(B) Uniform rectilinear motion
(C) Hyperbolic runaway motion
(D) Parabolic motion

82. The natural circular frequency omega_n of a mass m vibrating on a linear spring of stiffness k is:

(A) omega_n = sqrt(k / m)
(B) omega_n = sqrt(m / k)
(C) omega_n = k / m
(D) omega_n = 2 * pi * sqrt(k / m)

83. The natural period of oscillation T_n of the spring-mass oscillator in Q82 is given by:

(A) T_n = 2 * pi * sqrt(m / k)
(B) T_n = 2 * pi * sqrt(k / m)
(C) T_n = (1 / 2*pi) * sqrt(m/k)
(D) T_n = sqrt(m / k)

84. A particle's acceleration is velocity-dependent: a = - c * v (linear fluid viscous drag). The velocity decays exponentially as:

(A) v(t) = v0 * exp(- c * t)
(B) v(t) = v0 - c * t
(C) v(t) = v0 / (1 + c*t)
(D) v(t) = v0 * ln(c * t)

85. For the linear drag case in Q84 (a = - c * v), the maximum finite displacement s_max traversed as t approaches infinity is:

(A) s_max = v0 / c
(B) s_max = v0 * c
(C) s_max = v0^2 / (2 * c)
(D) Infinite

86. For quadratic aerodynamic drag (a = - k * v^2), terminal velocity v_term of a falling mass m under gravity g (drag force = k_drag * v^2) is:

(A) v_term = sqrt((m * g) / k_drag)
(B) v_term = (m * g) / k_drag
(C) v_term = sqrt(k_drag / (m * g))
(D) v_term = 0.5 * m * g / k_drag

87. Atwood's machine consists of two masses m1 = 6 kg and m2 = 4 kg connected by a light string over a frictionless pulley. The acceleration of the system is (g=9.81):

(A) 1.962 m/s^2 ((m1 - m2)/(m1 + m2) * g)
(B) 9.81 m/s^2
(C) 4.905 m/s^2
(D) 0.981 m/s^2

88. The tension T in the connecting cord of Atwood's machine in Q87 during motion is:

(A) 47.09 N (2*m1*m2/(m1+m2) * g)
(B) 58.86 N
(C) 39.24 N
(D) 98.10 N

89. A civil highway curve of radius R = 300 m is banked at angle theta to eliminate lateral tyre friction at design speed v = 108 km/h (30 m/s). The banking angle is (g=9.81):

(A) theta = atan(v^2 / (g * R)) = 17.02 degrees
(B) 10.5 degrees
(C) 25.4 degrees
(D) 30.0 degrees

90. On a flat unbanked highway curve of radius R = 150 m, the maximum vehicle speed without skidding given tyre friction mu_s = 0.60 is (g=9.81):

(A) 29.71 m/s (107.0 km/h)
(B) 20.50 m/s
(C) 15.00 m/s
(D) 35.20 m/s

91. A conical pendulum of mass m and cord length L revolves in a horizontal circle at constant angular velocity omega. The cord angle theta with the vertical satisfies:

(A) cos(theta) = g / (L * omega^2)
(B) sin(theta) = g / (L * omega^2)
(C) tan(theta) = L * omega^2 / g
(D) cos(theta) = L * omega^2 / g

92. A particle of mass m slides inside a smooth vertical circular hoop of radius R. The minimum velocity v_top required at the highest apex to maintain contact is:

(A) v_top = sqrt(g * R)
(B) v_top = sqrt(2 * g * R)
(C) v_top = sqrt(5 * g * R)
(D) v_top = 2 * sqrt(g * R)

93. For the particle in Q92, the corresponding minimum release velocity v_bottom at the lowest point of the hoop is:

(A) v_bottom = sqrt(5 * g * R)
(B) v_bottom = sqrt(4 * g * R)
(C) v_bottom = sqrt(3 * g * R)
(D) v_bottom = 3 * sqrt(g * R)

94. When the particle in Q93 passes through the lowest point with speed sqrt(5*g*R), the normal contact reaction exerted by the track is:

(A) N_bottom = 6 * m * g
(B) N_bottom = 5 * m * g
(C) N_bottom = m * g
(D) N_bottom = 4 * m * g

95. In oblique central impact of two smooth spheres, the component of velocity of each sphere that remains STRICTLY UNCHANGED during impact is:

(A) The velocity component tangential to the contact plane
(B) The velocity component normal to the contact plane
(C) The total speed of both spheres
(D) The relative velocity vector

96. The coefficient of restitution e in oblique impact is applied strictly to:

(A) The velocity components along the line of impact (common normal)
(B) The velocity components along the tangent line
(C) The resultant speeds
(D) The kinetic energies

97. The fraction of mechanical kinetic energy lost (Delta KE / KE_initial) in a perfectly plastic direct collision (e = 0) of equal masses (m1 = m2 = m, v2 = 0) is:

(A) 50 % (Delta KE = 0.5 * KE_initial)
(B) 100 %
(C) 25 %
(D) 75 %

98. A chain of length L and mass M hangs vertically from a ceiling. If released from rest, the force exerted on the floor when a length x has fallen and coiled is:

(A) F = 3 * (M / L) * g * x
(B) F = (M / L) * g * x
(C) F = 2 * (M / L) * g * x
(D) F = 0.5 * (M / L) * g * x

99. A variable-mass rocket of initial total mass m0 expels exhaust gases at constant relative speed u_rel at burn rate -dm/dt. Tsiolkovsky's rocket equation gives final speed:

(A) v = v0 + u_rel * ln(m0 / m)
(B) v = v0 + u_rel * (m0 / m)
(C) v = v0 + u_rel * (1 - m/m0)
(D) v = u_rel * exp(m0 / m)

100. A civil structural foundation block of mass M resting on a smooth surface is impacted by a seismic pulse represented by Dirac impulse F(t) = J0 * delta(t). The immediate velocity is:

(A) v(0+) = J0 / M
(B) v(0+) = J0 * M
(C) v(0+) = sqrt(2 * J0 / M)
(D) v(0+) = 0

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. State and explain the fundamental differential relations connecting position s(t), velocity v(t), and acceleration a(t) in rectilinear kinematics of particles.

Q2. Derive the three classical kinematic equations of motion for a particle travelling in a straight line with constant uniform acceleration 'a'.

Q3. A pile-driving hammer falls freely under gravity from rest at height h = 5.0 m onto a precast concrete pile foundation. Calculate: (1) Impact velocity v_impact, (2) If the pile penetrates Delta s = 0.08 m into soil before the hammer stops, compute the average deceleration and the soil

Q4. Explain the formulation of Curvilinear Motion in Normal-Tangential (n-t) coordinates. Derive expressions for tangential and normal acceleration components and state the physical interpretation of each.

Q5. A haulage dump truck travels along a circular horizontal haul road of radius rho = 250 m. Starting from rest at t = 0, its speed increases at constant rate a_t = 1.2 m/s^2. Determine: (1) Time required to achieve speed v = 18 m/s (64.8 km/h), (2) The normal acceleration a_n at that instant,

Q6. Derive the complete kinematic expressions for Projectile Motion in a vertical plane: (1) Equation of trajectory, (2) Time of flight T, (3) Maximum height H, (4) Horizontal range R.

Q7. State D'Alembert's Principle. Explain how it converts dynamic kinetics problems into equivalent static equilibrium systems, and demonstrate its formulation with a free-body diagram.

Q8. Two blocks A (mass m_A = 40 kg) and B (mass m_B = 25 kg) are connected by a light inextensible cable passing over a frictionless, massless pulley. Block A rests on a horizontal concrete slab with kinetic friction mu_k = 0.25, while Block B hangs freely. Using D'Alembert's Principle,

Q9. Derive the Work-Energy Theorem for a particle undergoing rectilinear translation subjected to a resultant force F(s).

Q10. Distinguish between Conservative and Non-Conservative forces in engineering mechanics. Derive the principle of Conservation of Mechanical Energy from the Work-Energy Theorem.

Q11. A civil railway buffer stop consists of two parallel coil springs of stiffness k = 150 kN/m each. A runaway shunting rail wagon of mass m = 18,000 kg strikes the buffer with an initial speed of v0 = 2.5 m/s. Using energy conservation, calculate: (1) Maximum compression x_max of the buffer

Q12. State the Principle of Linear Impulse and Momentum for a particle. Explain its derivation from Newton's Second Law and state under what physical conditions Linear Momentum is conserved.

Q13. A pile driving hammer of mass m1 = 1200 kg falls freely from height h = 3.0 m and strikes a steel pile of mass m2 = 600 kg. Assuming plastic impact (coefficient of restitution e = 0), calculate: (1) Velocity of hammer just before impact, (2) Common velocity of hammer and pile immediately

Q14. Define the Coefficient of Restitution (e) for collision of two bodies. Classify the impact regimes based on values of e and derive the formula for loss of kinetic energy in direct central impact.

Q15. A civil water tank overflow pipe discharges water horizontally from a height h = 20 m above ground at initial horizontal speed u = 12 m/s. Calculate: (1) Time taken for water stream to hit ground, (2) Horizontal distance x reached from pipe outlet, (3) Magnitude and angle of velocity at ground

Q16. Formulate the kinematics of a particle moving in a plane using Polar Coordinates (r, theta). Derive the equations for radial velocity, transverse velocity, radial acceleration, and transverse acceleration.

Q17. A crane boom rotates in a horizontal plane with constant angular speed theta_dot = 0.40 rad/s. Simultaneously, a trolley moves outward along the boom at constant relative speed r_dot = 1.50 m/s. When the trolley reaches radius r = 10.0 m, calculate: (1) Magnitude and direction of

Q18. Explain the civil engineering concept of Highway Superelevation (Banking of Roads). Derive the design relationship for equilibrium banking angle theta in terms of design speed v, curve radius R, and acceleration due to gravity g.

Q19. A civil highway curve of radius R = 200 m has a design speed limit of v = 72 km/h (20 m/s). (1) Determine the required equilibrium banking angle theta. (2) If a vehicle navigates the banked curve at v = 90 km/h (25 m/s), compute the minimum coefficient of static friction mu_s needed to

Q20. A continuous conveyor belt in a civil aggregates plant elevates gravel at mass flow rate dm/dt = 40 kg/s through vertical height h = 12 m at constant speed v = 2.5 m/s. Calculate: (1) Gravitational power required to elevate the gravel, (2) Kinematic power required to accelerate

Q21. Derive the relationship for the motion of a mass suspended from a linear spring executing undamped Simple Harmonic Motion (SHM). Determine the equation for natural circular frequency omega_n, natural frequency f_n, and time period T_n.

Q22. A 500 kg seismic testing block is supported on an elastomeric structural isolation bearing equivalent to a linear spring of stiffness k = 125 kN/m. Calculate: (1) Natural frequency f_n and period T_n, (2) If displaced by x0 = 20 mm and released from rest, write the displacement x(t) and

Q23. Analyze the direct central impact of two bodies A and B with masses m_A and m_B travelling with initial collinear velocities v_A1 and v_B1. Derive explicit formulas for post-impact velocities v_A2 and v_B2 in terms of coefficient of restitution 'e'.

Q24. A civil railway wagon A of mass m_A = 30 tonnes moves at v_A1 = 4.0 m/s and collides with stationary wagon B of mass m_B = 20 tonnes (v_B1 = 0). If the coefficient of restitution of the spring couplers is e = 0.60, calculate: (1) Velocities v_A2 and v_B2 after impact, (2) Total

Q25. Explain the mechanics of Oblique Central Impact between two smooth spherical particles. Detail the resolution along the Line of Impact (common normal) and the Plane of Contact (common tangent).

Q26. A 200 g tennis ball strikes a rigid concrete foundation slab at angle theta1 = 45 degrees to the surface normal with speed v1 = 20 m/s. If the coefficient of restitution between ball and slab is e = 0.70 (smooth slab), compute: (1) Rebound velocity v2 magnitude, (2) Rebound angle theta2 with

Q27. A civil construction safety net is designed to arrest a falling worker of mass m = 80 kg who falls from a bridge scaffold height h = 6.0 m. The safety net behaves as a non-linear spring absorbing kinetic energy with vertical deflection Delta y = 1.5 m before bringing the worker to rest.

Q28. Derive Tsiolkovsky's Rocket Equation for variable mass systems moving in free space without external gravity. Explain how Newton's Second Law is extended to systems with continuously changing mass.

Q29. A civil engineering cable car of mass M = 1500 kg is pulled up a 20-degree incline track by a winch cable. The winch motor applies a tractive force P(s) = (8000 + 400*s - 10*s^2) N, where s is distance along the track in meters. Track rolling resistance is constant at F_r = 1200 N. Using

Q30. Summarize the comparative methodologies for solving particle kinetics problems: (1) Direct Newton's Second Law / D'Alembert's Method, (2) Work-Energy Method, (3) Impulse-Momentum Method. Present guidelines for selecting the optimal method based on problem variables.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[A]	**Q21:**	[A]	**Q41:**	[A]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[A]	**Q22:**	[A]	**Q42:**	[A]	**Q62:**	[A]	**Q82:**	[A]
Q3:	[A]	**Q23:**	[A]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[A]	**Q24:**	[A]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[A]
Q5:	[A]	**Q25:**	[A]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[A]	**Q26:**	[A]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[A]	**Q27:**	[A]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[A]	**Q28:**	[A]	**Q48:**	[A]	**Q68:**	[A]	**Q88:**	[A]
Q9:	[A]	**Q29:**	[A]	**Q49:**	[A]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[A]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[A]	**Q31:**	[A]	**Q51:**	[A]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[A]	**Q32:**	[A]	**Q52:**	[A]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[A]	**Q33:**	[A]	**Q53:**	[A]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[A]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[A]	**Q35:**	[A]	**Q55:**	[A]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[A]	**Q36:**	[A]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[A]	**Q37:**	[A]	**Q57:**	[A]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[A]	**Q38:**	[A]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[A]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[A]	**Q40:**	[A]	**Q60:**	[A]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

Substituting into the acceleration equation:
a = (dv/dt) * u_t + (v^2 / rho) * u_n = a_t * u_t + a_n * u_n.
- Tangential Acceleration: a_t = dv/dt = v * (dv/ds). Physical meaning: Represents the time rate of change of the SPEED (magnitude of velocity).
- Normal Acceleration: a_n = v^2 / rho. Physical meaning: Represents the time rate of change of the DIRECTION of velocity (centripetal acceleration).
- Total Acceleration Magnitude: a = sqrt(a_t^2 + a_n^2).
- Direction Angle: phi = atan(a_n / a_t).

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g = T / 2.
H = y(t_apex) = u*sin(theta) * [(u*sin(theta))/g] - 0.5*g * [(u*sin(theta))/g]^2.
H = (u^2 * sin^2(theta)) / (2 * g).

5. Horizontal Range (R):
R = x(T) = u*cos(theta) * T = u*cos(theta) * [2*u*sin(theta) / g] = [u^2 *
(2*sin(theta)*cos(theta))] / g.
Using trigonometric identity 2*sin(theta)*cos(theta) = sin(2*theta):
R = (u^2 * sin(2*theta)) / g. Maximum range occurs when sin(2*theta) = 1 => 2*theta = 90 deg =>
theta = 45 degrees.

- Kinetic friction opposing motion: F_k = mu_k * N_A = 0.25 * 392.4 = 98.10 N (directed left).
- Cable tension T (directed right).
- D'Alembert inertia force: F_inertia_A = m_A * a = 40 * a (directed left, opposing motion).
Dynamic equilibrium equation (Sigma F_x = 0):
T - F_k - m_A * a = 0 => T - 98.10 - 40 * a = 0 => T = 40 * a + 98.10. (Equation 1)

3. Dynamic Equilibrium of Block B (Hanging Vertically):
Forces acting on B:
- Gravity weight: W_B = m_B * g = 25 * 9.81 = 245.25 N (downward).
- Cable tension T (upward).
- D'Alembert inertia force: F_inertia_B = m_B * a = 25 * a (upward, opposing downward acceleration).
Dynamic equilibrium equation (Sigma F_y = 0, downward positive):
W_B - T - m_B * a = 0 => 245.25 - T - 25 * a = 0 => T = 245.25 - 25 * a. (Equation 2)

4. Solve System Equations:
Equating Equation 1 and Equation 2:
40 * a + 98.10 = 245.25 - 25 * a => 65 * a = 147.15.
a = 147.15 / 65 = 2.2638 m/s^2 =~ 2.26 m/s^2.

5. Calculate Cable Tension T:
Substitute a into Eq 1: T = 40 * (2.2638) + 98.10 = 90.55 + 98.10 = 188.65 N.
Verify with Eq 2: T = 245.25 - 25 * (2.2638) = 245.25 - 56.60 = 188.65 N. (Exact match!)
Conclusion: System acceleration a = 2.26 m/s^2; Cable tension T = 188.65 N.

>>Solution to Question 9: Derive the Work-Energy Theorem for a particle undergoing rectilinear translation subjected to a resultant force F(s).

change in its kinetic energy.'
This scalar formulation eliminates the need to evaluate intermediate accelerations and time.

Final elastic potential energy: V_e2 = 0.5 * k_eq * x_max^2 = 0.5 * 300,000 * x_max^2 = 150,000 * x_max^2.

3. Solve for Maximum Compression x_max:
Conservation of mechanical energy (KE_1 + V_e1 = KE_2 + V_e2):
56,250 + 0 = 0 + 150,000 * x_max^2.
x_max^2 = 56,250 / 150,000 = 0.375 m^2.
x_max = sqrt(0.375) = 0.6124 m = 612.4 mm =~ 61.2 cm.

4. Maximum Retarding Force F_max:
F_max = k_eq * x_max = 300,000 N/m * 0.6124 m = 183,720 N = 183.72 kN.
Conclusion: Maximum spring compression x_max = 61.2 cm; Maximum retarding force = 183.72 kN.

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3. Loss of Kinetic Energy (Delta KE):
- Initial KE before impact: KE_initial = 0.5 * m1 * v1^2 = 0.5 * 1200 * (7.672)^2 = 600 * 58.86 = 35,316 J (35.32 kJ).
- Final KE after impact: KE_final = 0.5 * (m1 + m2) * V_common^2 = 0.5 * 1800 * (5.115)^2 = 900 * 26.16 = 23,544 J (23.54 kJ).
- Energy lost: Delta KE = KE_initial - KE_final = 35,316 - 23,544 = 11,772 J (11.77 kJ).
- Percentage energy lost: (11,772 / 35,316) * 100% = 33.33%.
Note: For plastic impact of masses m1 and m2 with m2 initially at rest, fractional loss is exactly m2 / (m1 + m2) = 600 / 1800 = 1/3 = 33.33%.

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3. Impact Velocity Components and Magnitude:
- Horizontal velocity component at impact: v_x = u = 12.0 m/s.
- Vertical velocity component at impact: v_y = g * t_flight = 9.81 * 2.019 = 19.809 m/s (downward).
- Total impact speed: v = sqrt(v_x^2 + v_y^2) = sqrt((12.0)^2 + (19.809)^2) = sqrt(144 + 392.40) = sqrt(536.40) = 23.16 m/s.

4. Impact Angle theta with Ground Horizontal:
tan(theta) = v_y / v_x = 19.809 / 12.0 = 1.6508 => theta = atan(1.6508) = 58.79 degrees below horizontal.
Conclusion: Time of flight = 2.02 s; Horizontal distance = 24.23 m; Impact speed = 23.16 m/s at 58.8 deg.

[Detailed Step-by-Step Answer]:
1. Given Kinematic Data:
r = 10.0 m; r_dot = 1.50 m/s; r_ddot = 0 (constant extension speed).
theta_dot = 0.40 rad/s; theta_ddot = 0 (constant angular speed).

2. Velocity Components and Magnitude:
v_r = r_dot = 1.50 m/s.
v_theta = r * theta_dot = 10.0 * 0.40 = 4.00 m/s.
v = sqrt(v_r^2 + v_theta^2) = sqrt((1.50)^2 + (4.00)^2) = sqrt(2.25 + 16.00) = sqrt(18.25) = 4.272 m/s =~ 4.27 m/s.

3. Acceleration Components:
- Radial acceleration a_r = r_ddot - r * (theta_dot)^2:
a_r = 0 - 10.0 * (0.40)^2 = - 10.0 * 0.16 = - 1.60 m/s^2 (directed radially inward).
- Transverse acceleration a_theta = r * theta_ddot + 2 * r_dot * theta_dot:
a_theta = 0 + 2 * (1.50) * (0.40) = 1.20 m/s^2 (Coriolis acceleration contribution).

4. Total Acceleration Magnitude:
a = sqrt(a_r^2 + a_theta^2) = sqrt((- 1.60)^2 + (1.20)^2) = sqrt(2.56 + 1.44) = sqrt(4.00) = 2.00 m/s^2.
Direction angle beta with radial vector: tan(beta) = a_theta / |a_r| = 1.20 / 1.60 = 0.75 => beta = 36.87 degrees.
Conclusion: Velocity v = 4.27 m/s; Acceleration a = 2.00 m/s^2.

>> Solution to Question 18

- Equilibrium banking angle: theta = atan(v^2 / (g * R)).
- At this exact design speed, lateral tyre friction force is IDENTICALLY ZERO, maximizing passenger comfort and pavement durability.

Conclusion: Gravity power = 4.71 kW; Acceleration power = 0.25 kW; Motor power required = 5.83 kW.

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Delta KE = KE_1 - KE_2 = 240,000 - 178,560 = 61,440 J (61.44 kJ).
Percentage Loss = (61.44 / 240.0) * 100% = 25.60%.
Theoretical check: 0.5 * (30*20/50)*1000 * (4.0)^2 * (1 - 0.6^2) = 0.5 * 12000 * 16 * 0.64 = 61,440 J. (Exact!)
Conclusion: v_A2 = 1.44 m/s; v_B2 = 3.84 m/s; Energy lost = 61.44 kJ (25.6%).

- Angle with surface normal: tan(theta2) = v_t2 / v_n2 = 14.142 / 9.8995 = 1.4286 => theta2 = atan(1.4286) = 55.01 degrees.

4. Linear Impulse Exerted on Ball:
All impulse is normal: J_n = m * [v_n2 - (- v_n1)] = m * (v_n2 + v_n1).
J_n = 0.200 kg * (9.8995 + 14.142) = 0.200 * 24.0415 = 4.808 N*s.
Conclusion: Rebound speed = 17.26 m/s; Rebound angle with normal = 55.01 degrees; Normal impulse = 4.81 N*s.

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m * dv = - u_e * dm => dv = - u_e * (dm / m).

4. Integration Over Burn Duration:
Integrating from initial launch mass m0 to burnout mass m_f:
Integral_v0^v dv = - u_e * Integral_m0^mf (dm / m) = - u_e * [ln(mf) - ln(m0)].
v - v0 = u_e * ln(m0 / mf).
Tsiolkovsky's Rocket Equation: Delta v = u_e * ln(m0 / mf).
This fundamental result demonstrates that velocity gain depends logarithmically on mass ratio m0/mf and linearly on exhaust velocity.

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* Integrates forces over spatial displacement: U = Integral(F ds).
* Eliminates internal normal reactions and frictionless geometric constraint forces.

- Method 3: Impulse-Momentum Principle:
* Governing equation: Integral_t1^t2 Sigma F dt = m * v2 - m * v1 = Delta p.
* Vector formulation: Integrates forces over elapsed TIME interval dt.
* Ideal for transient high-magnitude forces (impact, explosions, pile-driving) where contact time dt -> 0.

2. Engineering Decision Matrix / Selection Strategy:
- Select Work-Energy when variables involve Force, Distance (Displacement), and Velocity (F, s, v), and time 't' is neither given nor required.
- Select Impulse-Momentum when variables involve Force, Time, and Velocity (F, t, v), or when solving collisions and impact events.
- Select Newton's Second Law / D'Alembert when normal forces, friction limits, instantaneous cable tensions, or accelerations (a) are explicitly sought.
